

MULTI-LLM BENCHMARKING & EVALUATION PLATFORM

Technical Documentation & API Reference Manual

1. SYSTEM OVERVIEW & ARCHITECTURE

The Multi-LLM Benchmarking & Evaluation Platform is an enterprise-grade Django REST API application designed to dynamically process user prompts, evaluate AI model outputs through a double-blind jury system, and resolve a definitive winner utilizing a statistical consensus-scoring mechanism.

Core Architectural Pipeline:

1. **Ingress Validation:** The client submits a payload containing a prompt. The platform filters and validates the input length via serialization constraints.
2. **Asynchronous Orchestration:** Using Python's `asyncio` engine, the backend dispatches non-blocking concurrent API requests to multiple upstream LLM providers (OpenAI, Gemini, DeepSeek, Groq).
3. **Double-Blind Anonymization:** Model identities are stripped dynamically and mapped to uppercase token labels (`Response A`, `Response B`, etc.) to guarantee unbiased downstream evaluation.
4. **Independent Evaluation Jury:** Anonymized outputs are parsed through multi-agent AI Judges (Gemini and Groq) evaluating core linguistic vectors: Accuracy, Relevance, Completeness, Clarity, and Conciseness.
5. **Statistical Consensus Blending :** Text structures undergo TF-IDF vectorization and Cosine Similarity checking to isolate aberrant hallucinations or outliers, updating model scores mathematically before crowning the session winner.
6. **Relational Persistence:** Complete session audit logs are atomically saved inside a relational storage engine (PostgreSQL) using optimized query techniques.

2. ENVIRONMENT CONFIGURATION (`.env`)

To run this application locally or in production, create a `.env` file in the root project directory. Copy the sample block below and replace the placeholders with your actual API gateway credentials.

```
# Server Deployment Configuration
```

```
DEBUG=True
```

```
SECRET_KEY=django-insecure-mock-key-evaluation-platform-2026
```

Upstream LLM Provider API Keys

OPENAI_API_KEY=sk-proj-mockOpenAiKeyForTestingPurposesOnly12345

GEMINI_API_KEY=AlzaSyMockGeminiKeyGenAIPlatformRegistration2026

DEEPSEEK_API_KEY=sk-mockDeepSeekOperationalKeyGatewayAccessX7

GROQ_API_KEY=gsk_mockGroqHighPerformanceInferenceEngineKey99

Database Configuration (Uncomment for PostgreSQL Setup)

DB_NAME=multi_llm_bench_db

DB_USER=postgres

DB_PASSWORD=your_secure_db_password

DB_HOST=localhost

DB_PORT=5432\

3. API SPECIFICATIONS & DOCUMENTATION

ENDPOINT 1: Submit Prompt & Execute Evaluation Benchmark

- **URL:** `/api/evaluate/`
- **Method:** `POST`
- **Headers:** `Content-Type: application/json`

http://127.0.0.1:8000/api/evaluate/

Server response

Code Details

200
Undocumented

Response body

```
{
  "session_id": 1,
  "prompt": "what is blackhole",
  "winner": "OpenRouter-gpt-oss-20b-free",
  "final_score": 9.5,
  "all_responses": [
    {
      "label": "A",
      "model_name": "Groq-llama-3.1-8b-instant",
      "response_text": "A black hole is a region in space where the gravitational pull is so strong that nothing, including light, can escape. It is formed when a massive star collapses in on itself and its gravity becomes so strong that it warps the fabric of spacetime around it.\n\nHere's a simplified explanation of how a black hole is formed:\n1. **Massive Star Collapse**: A massive star runs out of fuel and dies. Its core collapses under its own gravity, causing a massive amount of matter to be compressed into an incredibly small space.\n2. **Singularity Formation**: The collapse creates a point of infinite density and zero volume, known as a singularity. This is the center of the black hole.\n3. **Event Horizon Formation**: As matter falls towards the singularity, it creates a boundary called the event horizon. Once something crosses the event horizon, it is trapped by the black hole's gravity and cannot escape.\n4. **Gravitational Pull**: The event horizon marks the point of no return. Any matter or radiation that enters the event horizon is pulled towards the singularity, never to escape.\n\n**Characteristics of Black Holes**:\n1. **Mass**: Black holes have mass, which determines the strength of their gravity.\n2. **Spin**: Black holes can rotate, which affects the direction of their jets of energy and the way they interact with their surroundings.\n3. **Charge**: Black holes can have an electric charge, which affects their behavior in the presence of other charged objects.\n\n**Types of Black Holes**:\n1. **Stellar Black Holes**: Formed from the collapse of individual stars, these black holes have masses up to about 20 times that of the sun.\n2. **Supermassive Black Holes**: Found at the centers of galaxies, these black holes have masses millions or even billions of times that of the sun.\n3. **Intermediate-Mass Black Holes**: Black holes with masses between those of stellar and supermassive black holes.\n\n**Observational Evidence for Black Holes**:\n1. **X-rays and Gamma Rays**: Telescopes can detect X-rays and gamma rays emitted by hot gas swirling around black holes.\n2. **Radio Waves**: Radio telescopes can detect radio waves emitted by matter as it spirals into a black hole.\n3. **Gravitational Waves**: The detection of gravitational waves by LIGO and VIRGO provide strong evidence for the existence of black holes.\n\n**Interesting Facts about Black Holes**:\n1. **Information Paradox**: The laws of quantum mechanics suggest that information cannot be destroyed, but the laws of general relativity suggest that information that falls into a black hole is lost forever.\n2. **Black Hole Entropy**: Black holes have a temperature and entropy, which is related to the area of their event horizon.\n3. **Black Hole Complementarity**: The information paradox is resolved by considering that information that falls into a black hole is not lost, but is instead stored on the event horizon."
    }
  ]
}
```

Response headers

Without addition of consensus

POST http://127.0.0.1:8000/api/evaluate/ Send

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Headers 8 hidden

Key	Value	Description	Bulk Edit	Presets	...
Content-Type	application/json				

Body Cookies Headers 10 Test Results 200 OK • 15.41 s • 5.05 KB

{ } JSON Preview Visualize

```
1 {
2   "session_id": 2,
3   "prompt": "briefly tell what is machine learning?",
4   "winner": "Groq-llama-3.1-8b-instant",
5   "final_score": 9.0,
6   "all_responses": [
7     {
8       "label": "A",
9       "model_name": "Groq-llama-3.1-8b-instant",
10      "response_text": "**Machine Learning (ML)** is a subset of artificial intelligence (AI) that involves training algorithms",
11      "response_time_ms": 1085.81,
12      "groq_score": 9,
13      "gemini_score": 9.0,
14      "final_score": 9.0
15    },
16    {
17      "label": "B",
18      "model_name": "Groq-qwen/qwen3-32b",
19      "response_text": "Machine Learning (ML) is a subset of artificial intelligence (AI) that involves training algorithms",
20      "response_time_ms": 1085.81,
21      "groq_score": 9,
22      "gemini_score": 9.0,
23      "final_score": 9.0
24    }
25  ]
26 }
```

With consensus

The screenshot shows a REST client interface with a POST request to `http://127.0.0.1:8000/api/evaluate/`. The response is a 200 OK status with a JSON body. The JSON body contains the following data:

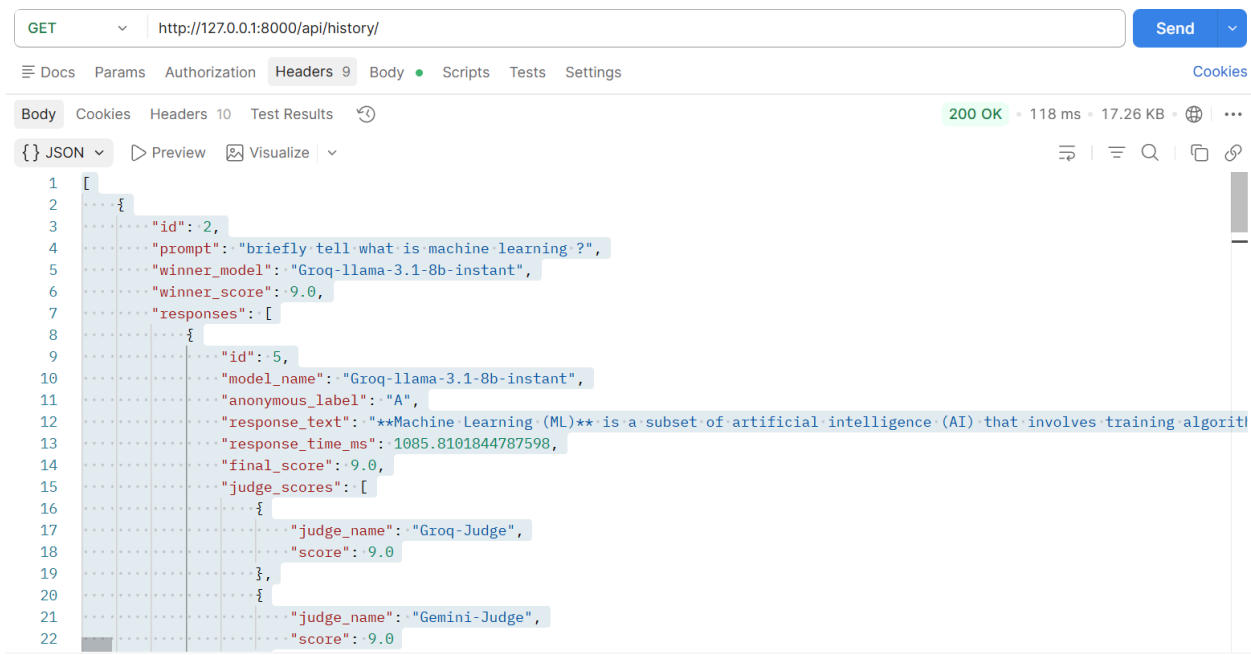
```
{
  "session_id": 6,
  "prompt": "AI KYA HA ?",
  "winner": "OpenRouter-gpt-oss-20b:free",
  "final_score": 6.98,
  "judges_used": {
    "groq": true,
    "gemini": true
  },
  "consensus": {
    "scores": {
      "A": 0.0954,
      "B": 0.1083,
      "C": 0.1022,
      "D": 0.0916
    },
    "outliers": [],
    "confidence": "low"
  }
}
```

ENDPOINT 2: Historical Evaluation Sessions Log

- **URL:** `/api/history/`
- **Method:** `GET`
- **Headers:** `Accept: application/json`
- **Description:** Retrieves a registry containing all historical model comparison evaluation records. The ORM query layers execute structural `prefetch_related` wrappers to eliminate database performance bottlenecks (N+1 query loops).

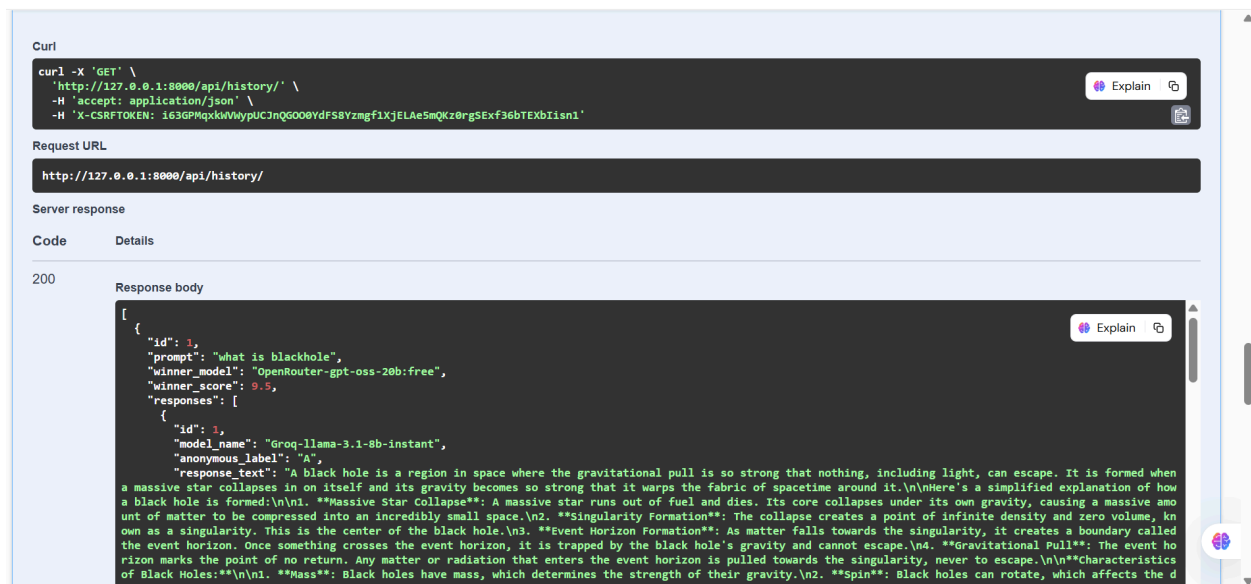
The screenshot shows a REST client interface with a GET request to `http://127.0.0.1:8000/api/history/`. The response is a 200 status with a JSON body. The JSON body contains the following data:

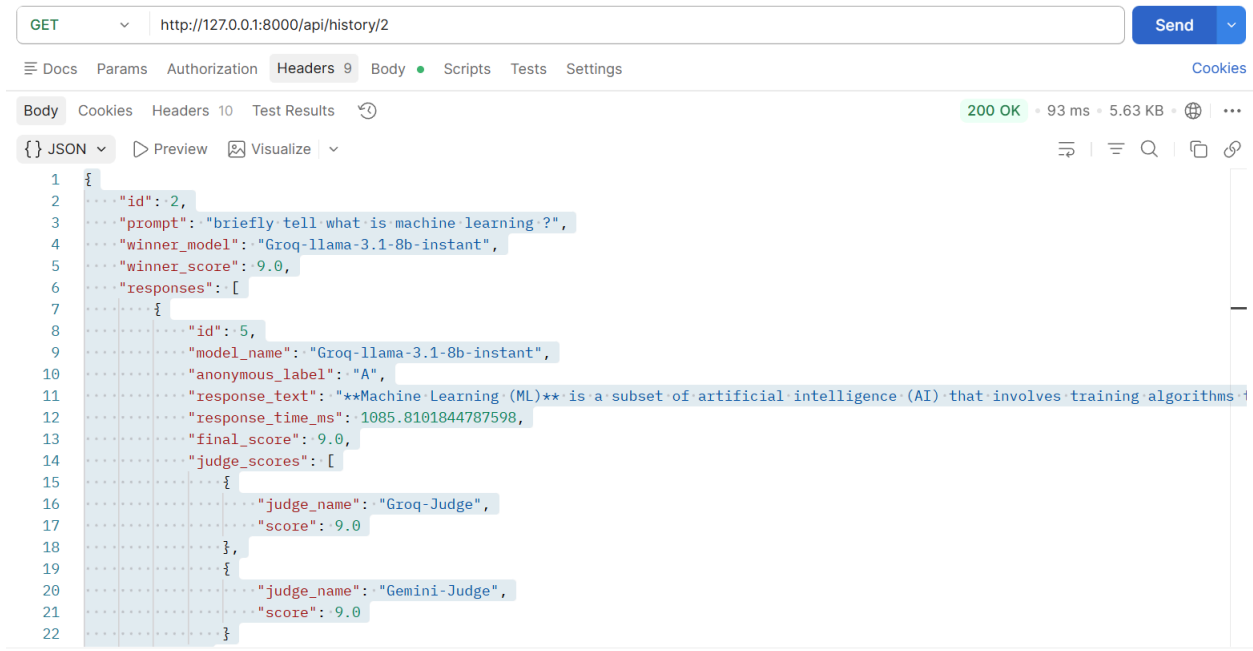
```
{
  "id": 1,
  "prompt": "what is blackhole",
  "winner_model": "OpenRouter-gpt-oss-20b:free",
  "winner_score": 9.5,
  "responses": [
    {
      "id": 1,
      "model_name": "Groq-llama-3.1-8b-instant",
      "anonymous_label": "A",
      "response_text": "A black hole is a region in space where the gravitational pull is so strong that nothing, including light, can escape. It is formed when a massive star collapses in on itself and its gravity becomes so strong that it warps the fabric of spacetime around it. Here's a simplified explanation of how a black hole is formed:\n1. **Massive Star Collapse**: A massive star runs out of fuel and dies. Its core collapses under its own gravity, causing a massive amount of matter to be compressed into an incredibly small space.\n2. **Singularity Formation**: The collapse creates a point of infinite density and zero volume, known as a singularity. This is the center of the black hole.\n3. **Event Horizon Formation**: As matter falls towards the singularity, it creates a boundary called the event horizon. Once something crosses the event horizon, it is trapped by the black hole's gravity and cannot escape.\n4. **Gravitational Pull**: The event horizon marks the point of no return. Any matter or radiation that enters the event horizon is pulled towards the singularity, never to escape.\n\n**Characteristics of Black Holes**:\n1. **Mass**: Black holes have mass, which determines the strength of their gravity.\n2. **Spin**: Black holes can rotate, which affects the distribution of their spacetime curvature around them.\n3. **Charge**: Black holes can have an electric charge, which affects their interaction with other charged particles."
    }
  ]
}
```



ENDPOINT 3: Isolated Session Details

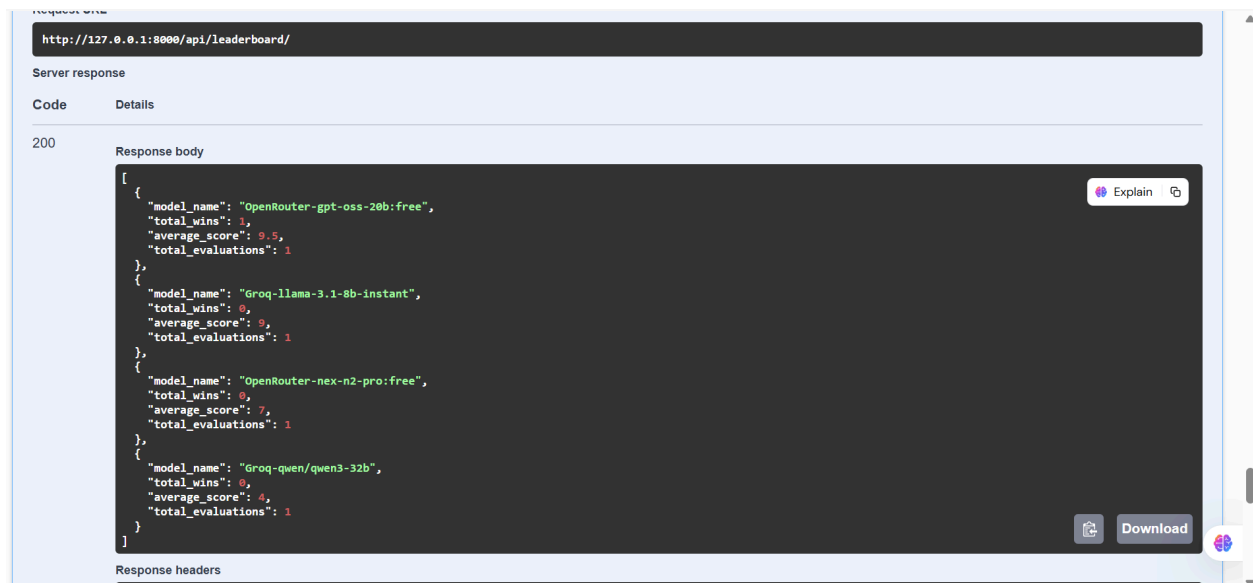
- **URL:** `/api/history/<id>/` (e.g., `/api/history/1/`)
- **Method:** `GET`
- **Headers:** `Accept: application/json`
- **Description:** Pulls isolated deep audit logs for a single target execution session parsed via its integer primary key ID. Validates route data types, rendering automatic `404 Not Found` messages if indices do not exist.

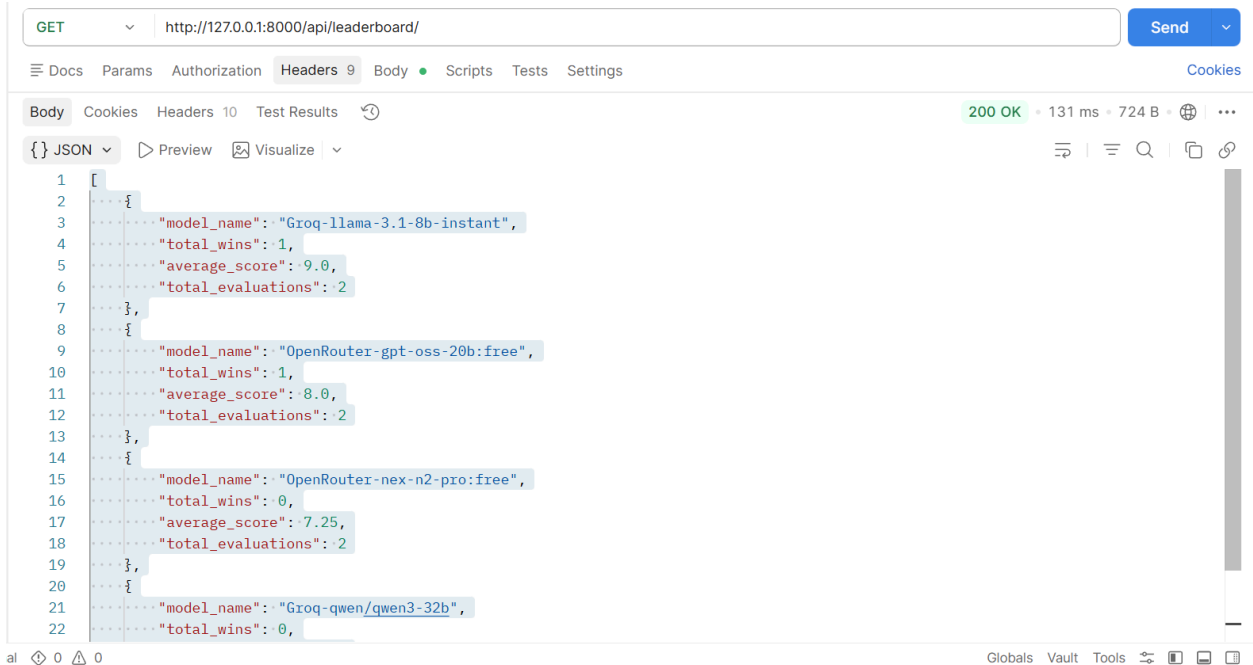




ENDPOINT 4: Global Performance Leaderboard Rankings

- **URL:** `/api/leaderboard/`
- **Method:** `GET`
- **Headers:** `Accept: application/json`
- **Description:** Compiles real-time analytical metrics by grouping historical evaluation blocks at the database layer. Leverages Django `Count` and `Avg` functions to compute global model performance stats, ordering the output dynamically using total match wins.





4. ADVANCED FEATURE OVERVIEW: STATISTICAL CONSENSUS ENGINE

To meet the evaluation **Bonus Challenge** criteria, the platform implements an embedded mathematical tracking ecosystem inside `consensus.py` instead of relying blindly on simple judge score averages.

- **Semantic Similarity Vectoring:** Generated model response texts are parsed dynamically using TF-IDF (Term Frequency-Inverse Document Frequency) matrices. The system then calculates cross-token agreement distributions using Cosine Similarity checks.
- **Algorithmic Outlier Isolation:** Any model response displaying extreme deviations from structural similarity averages (Standard Deviation boundary factors exceeding $1.5 \times \sigma$ combined with raw similarity margins falling below 0.15) is flagged as an aberrant outlier or hallucination.
- **Weighted Score Penalty Mitigation:** When an outlier node is isolated, its judge criteria calculations are dynamically down-weighted to minimize its chance of being selected as the winner, preventing corrupted tokens from gaming the leaderboard rankings.

$$\text{Adjusted Score} = (\text{Raw Judge Score} \times (1 - w)) + (\text{Consensus Scale} \times w)$$

```
Adjusted Score = (Raw Judge Score x (1 - w)) + (Consensus Scale x w)
```

```
Where:  w = 0.20  (consensus weight = 20%)
        1 - w = 0.80  (judge weight = 80%)
```

Example:

```
Judge Score    = 8.5
Consensus      = 0.91 --> Scaled = 0.91 x 10 = 9.1
Adjusted       = (8.5 x 0.80) + (9.1 x 0.20)
```

```
= 6.80 + 1.82
= 8.62 <-- Final Score
```

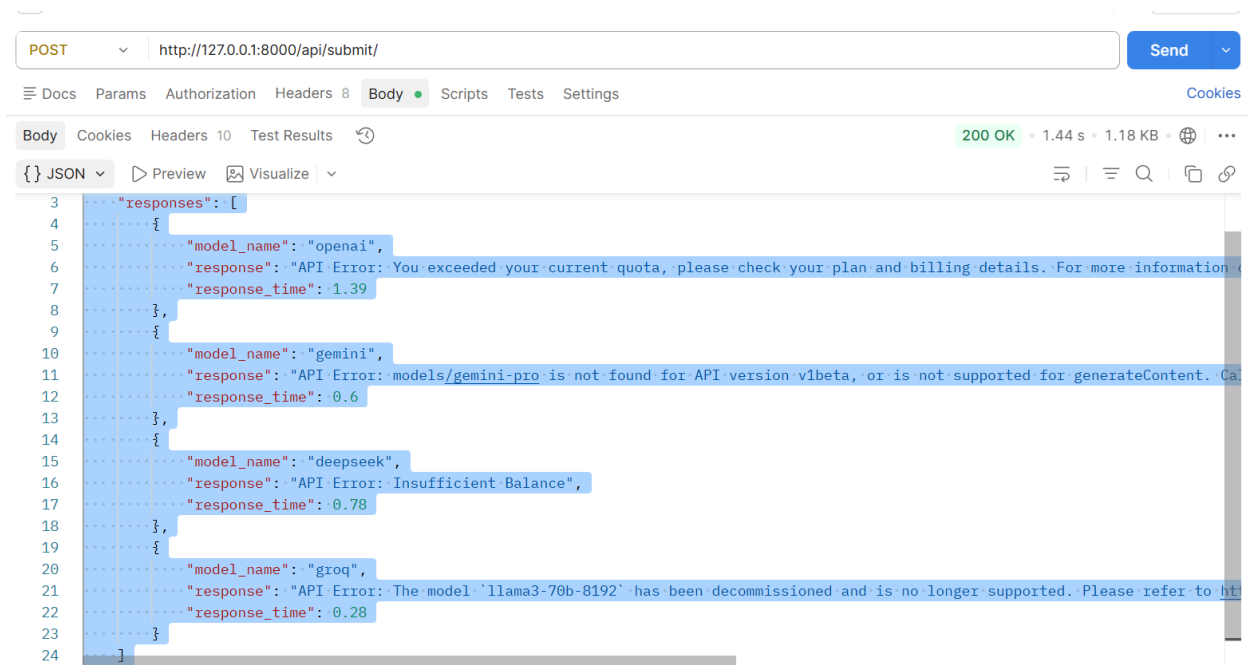
Challenges Faced and Resolutions

1. OpenAI API Quota Exceeded

Issue: You exceeded your current quota, please check your plan and billing details.

Cause: The API key was valid, but the OpenAI account had no available quota or active billing.

Resolution: OpenAI integration done using openrouter api.



2. Gemini Model Not Found

Issue: models/gemini-1.5-flash is not found for API version v1b

Cause: The application was using an outdated Gemini model that was no longer available for the provided API version.

Resolution: Available models were retrieved using the Gemini Models API and the integration was updated to use gemini-2.5-flash

3. Gemini Free Tier Quota Exceeded

Issue:Quota exceeded for metric generate_content_free_tier_requests

Cause:the Gemini free-tier request limit had been reached.

Resolution:Retry handling was considered and alternative judge models were used when quota limits were encountered.

4. DeepSeek Insufficient Balance

POST

▼

https://api.deepseek.com/chat/completions

☰ Docs

Params

Authorization

Headers 9

Body ●

Scripts

Tests

Se

Query Params

Key	Value
Key	Value

Body

Cookies

Headers 14

Test Results

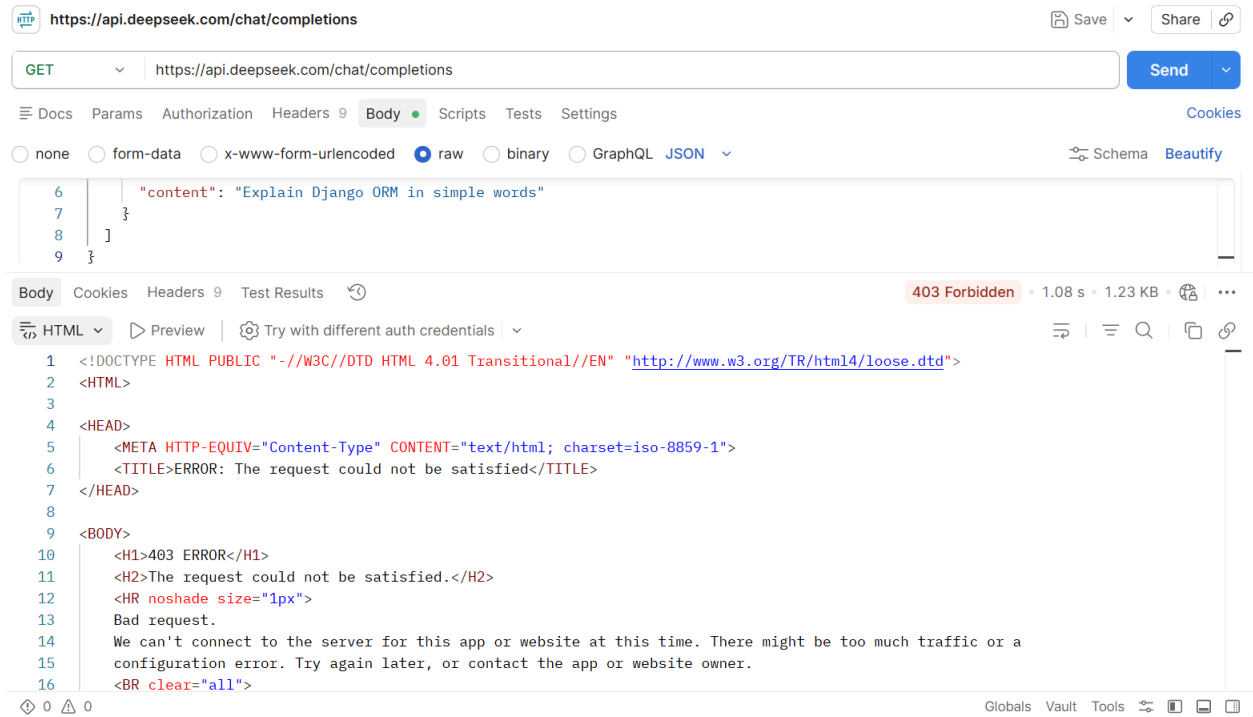
🔄

{ } JSON ▼

▶ Preview

🐞 Debug with AI ▼

```
1  {
2    "error": {
3      "message": "Insufficient Balance",
4      "type": "unknown_error",
5      "param": null,
6      "code": "invalid_request_error"
7    }
8  }
```



Issue:Insufficient Balance

Cause:The DeepSeek account did not have sufficient credits to process requests.

Resolution:DeepSeek was removed from active evaluation and replaced with Groq-hosted models.

5. Groq Model Deprecation

Issue:The model llama-3.1-70b-versatile has been decommissioned

Cause:The selected Groq model had been officially deprecated.

Resolution:The deprecated model was replaced with supported models:

llama-3.1-8b-instant

qwen/qwen3-32b

openai/gpt-oss-20b

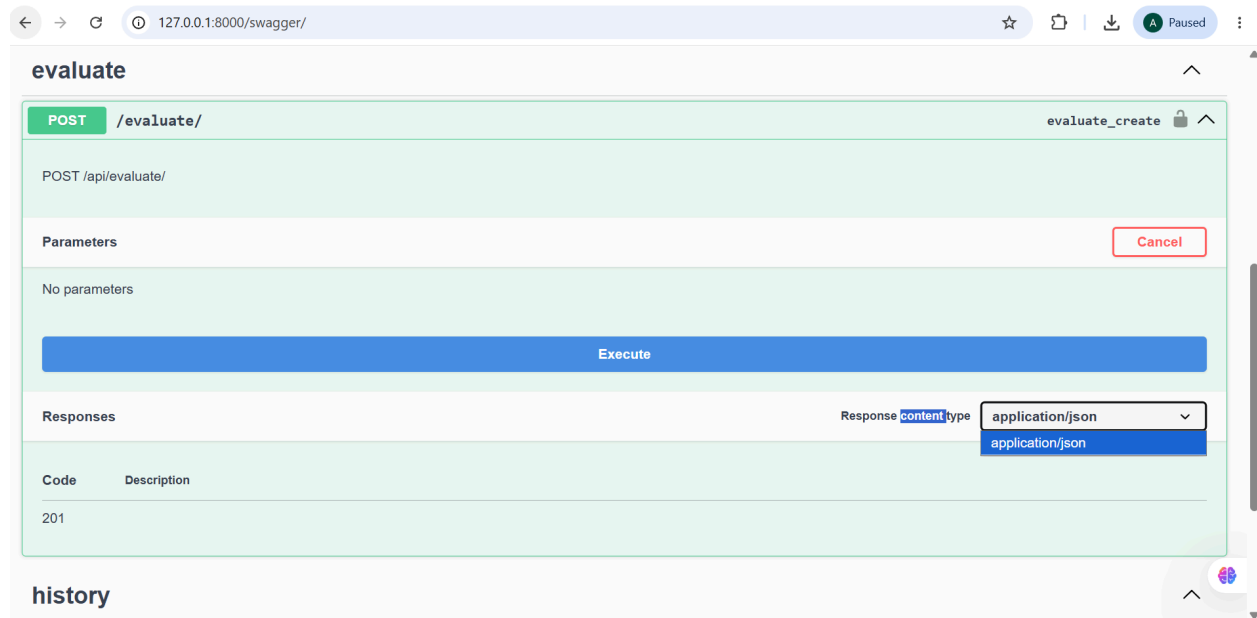
6. Django Async ORM Error

Issue:SynchronousOnlyOperation:You cannot call this from an async context

Cause: Django ORM queries were executed directly inside asynchronous functions.

Resolution: Database operations were moved outside async contexts or wrapped using asynchronous-safe techniques.

7. Swagger UI Schema Mapping Issue



Error: Swagger UI displayed "No parameters" for the `/api/evaluate/` POST endpoint.

Cause: The endpoint was powered by a standard Django REST Framework `APIView`. By default, `drf-yasg` (Swagger generator) cannot auto-detect the expected input data payload structure for raw `APIView` classes unless explicitly documented. Because there was no explicit link between the view and the data validator (`PromptInputSerializer`), Swagger rendered an empty parameter block.

Resolution: Decorated the view function with `@swagger_auto_schema(request_body=PromptInputSerializer)`. This explicitly mapped the serializer schema to the Swagger engine, turning the static documentation into an interactive JSON input portal with an "Edit Value" field.

127.0.0.1:8000/swagger/

POST /evaluate/ evaluate_create

POST /api/evaluate/

Parameters Cancel

Name	Description
data required	
object	Edit Value Model
(body)	

PromptInput {
 prompt*
 }

Prompt string
 title: Prompt
 maxLength: 2000
 minLength: 5

Execute

Responses Response content type: application/json

8.hugging face api did not worked

https://api-inference.huggingface.co/models/meta-llama/Llama-3.2-3B-Instruct/v1/chat/completions Save Share

POST https://api-inference.huggingface.co/models/meta-llama/Llama-3.2-3B-Instruct/v1/chat/completions Send

Docs Params Authorization Headers 9 Body Scripts Tests Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```

1 {
2   "messages": [
3     {
4       "role": "user",
5       "content": "Explain Django ORM in simple words"
6     }
7   ]
8 }

```

Response

Could not send request

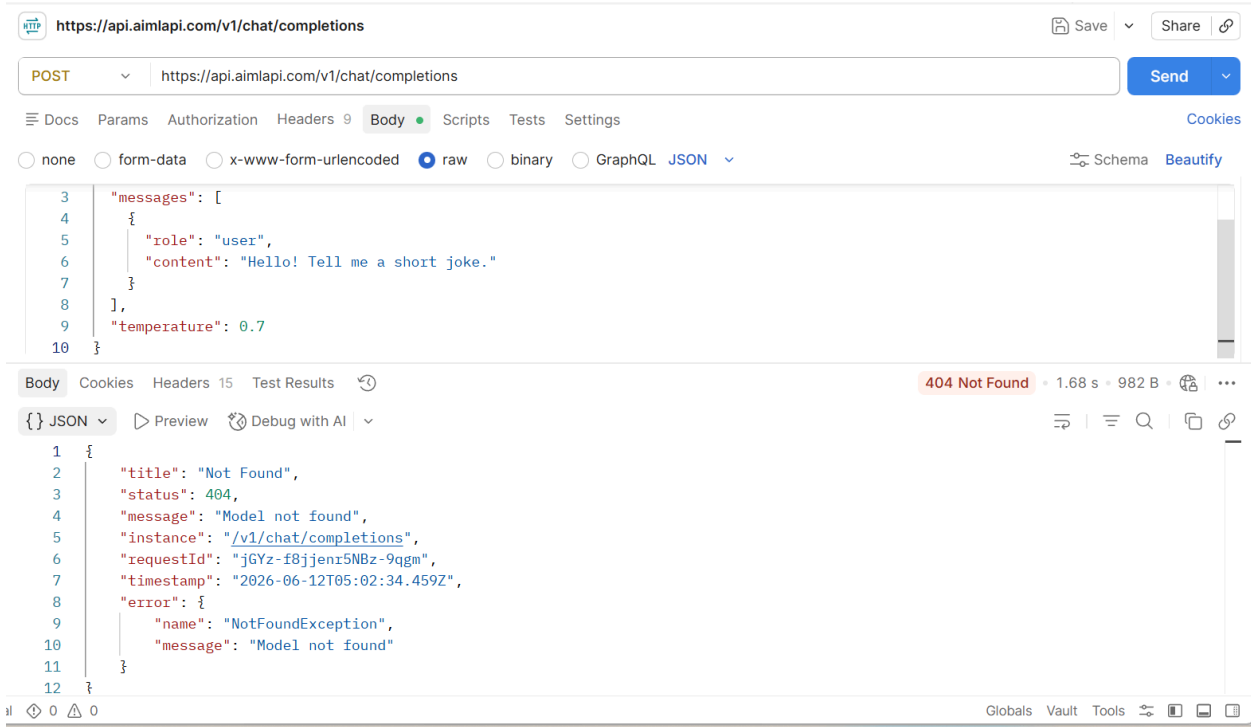
Error: getaddrinfo ENOTFOUND api-inference.huggingface.co Debug with AI

[View in console](#) • [Learn about troubleshooting requests](#)

al 0 0 Globals Vault Tools

getaddrinfo ENOTFOUND api-inference.huggingface.co — Could not send request.

9.aiml api did not work



404 Not Found — Model not found. NotFoundException at /v1/chat/completions.

Final Integrated Models

LLM Models (Response Generation)

Model	Provider	API	Label
llama-3.1-8b-instant	Groq	api.groq.com	A
qwen/qwen3-32b	Groq	api.groq.com	B
nex-agi/nex-n2-pro:free	OpenRouter	openrouter.ai	C
gpt-oss-20b:free	OpenRouter	openrouter.ai	D

Judge Models (Evaluation)

Judge	Model Used	Scoring Criteria
Groq Judge	llama-3.1-8b-instant	Accuracy, Relevance, Completeness, Clarity, Conciseness

Gemini Judge	gemini-2.5-flash	Accuracy, Relevance, Completeness, Clarity, Conciseness
---------------------	------------------	---

Both judges independently evaluate anonymized model responses (labeled A, B, C, D). Final score = Average(Groq Judge Score, Gemini Judge Score). Winner = highest adjusted score after consensus blending.